Pimpri Chinchwad University

Department of Computer Science and Engineering (AI & ML)

Python Programming: Mini Project Report

RENT CALCULATOR

A Python program to split rent, food and electricity costs among room/flat mates

Submitted by

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Class / Year	Second Year
Subject	Python Programming
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College	Pimpri Chinchwad University
Academic Year	2026-27

Submitted to the Department as part of the Python Programming course work

1. Problem Statement

Students and working people who share a hostel room or flat have to divide several monthly costs among themselves: the room rent, food ordered together (for example snacks), and the electricity bill. The electricity bill is not a fixed amount. It depends on the units consumed and the charge per unit, so the share of each person has to be calculated every month.

Doing this by hand takes time, and a small mistake in the arithmetic can lead to arguments between room-mates. This project solves the problem with a short Python program. The user types in the costs once, and the program calculates the electricity bill, adds all the costs and prints how much each person has to pay.

2. Objectives

- To take the rent, food expense, electricity units, charge per unit and number of persons as input from the user.
- To calculate the electricity bill from the units consumed and the charge per unit.
- To calculate the total expense (rent + food + electricity) and divide it equally among all persons.
- To display the amount each person has to pay in a clear message.
- To practise Python basics: variables, input/output, type conversion and arithmetic operators.

3. Technologies Used

Item	Details
Programming language	Python 3
Environment	Jupyter Notebook (the output screenshot is from a notebook cell); source file saved as Rent_calculator.py
Libraries / modules	None. Only Python's built-in functions are used.
Built-in functions	input() to read values, int() to convert text to whole numbers, print() to show the result
Operators	Multiplication (*), addition (+), floor division (//)

4. Project Approach / Methodology

4.1 Inputs, Processing and Output

Stage	Description
Input	rent, food (amount spent on food ordered), electricity_spend (electricity units consumed), charge_per_unit, persons (number of people sharing)
Processing	total_bill = electricity_spend * charge_per_unit; output = (food + rent + total_bill) // persons
Output	The message "Each person will pay =" followed by the share of each person

4.2 Working Steps and Flowchart

1. The program starts and asks the user for the hostel/flat rent.
2. It then asks for the food amount, the electricity units, the charge per unit and the number of persons. Each value is converted to an integer using int().
3. The electricity bill is calculated: total_bill = electricity_spend * charge_per_unit.
4. The three costs are added and divided by the number of persons using floor division (//), so the share is a whole number.
5. The result is printed as "Each person will pay =" and the program ends.

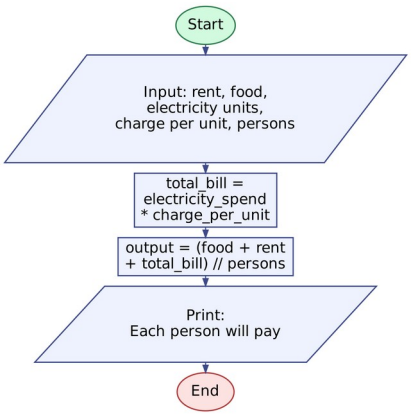


Fig. 1: Flowchart of the program

5. Implementation

The whole program is written as a short sequence of statements. It does not define any user-defined functions or use any external library. The important sections are explained below.

5.1 Taking input

```
rent = int(input("Enter your hostel/flat rent = "))
food = int(input("Enter the amount of food ordered = "))
electricity_spend = int(input("Enter the total of electricity spend = "))
charge_per_unit = int(input("Enter the charge per unit = "))
persons = int(input("Enter the number of persons living in room/flat = "))
```

input() reads what the user types as text. int() converts that text into a whole number so that it can be used in calculations. The five values are stored in the variables rent, food, electricity_spend, charge_per_unit and persons.

5.2 Calculation

```
total_bill = electricity_spend * charge_per_unit
output = (food + rent + total_bill) // persons
```

The first line finds the electricity bill by multiplying the units consumed with the charge per unit. The second line adds the food expense, the rent and the electricity bill, and divides the total by the number of persons. The operator // is floor division, so the answer is a whole number and any remainder is dropped.

5.3 Displaying the result

```
print("Each person will pay = ", output)
```

print() shows the final share of each person on the screen.

6. Output Screenshot and Testing

Figure 2 shows the output of the program when it was run in a Jupyter Notebook cell. The values entered by the user are rent = 10000, food = 4000, electricity units = 2000, charge per unit = 8 and persons = 4.

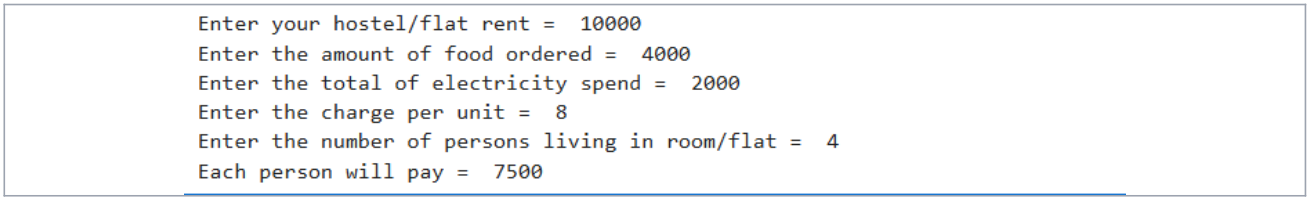


Fig. 2: Program output in Jupyter Notebook

Explanation: the electricity bill is 2000 × 8 = 16000. The total expense is 4000 + 10000 + 16000 = 30000. Dividing by 4 persons gives 7500, which matches the program output "Each person will pay = 7500".

Test cases

No.	Inputs (rent, food, units, rate, persons)	Expected result	Actual result	Source of result
1	10000, 4000, 2000, 8, 4	7500	7500 (matches)	Screenshot (Fig. 2)
2	6000, 1500, 100, 7, 3	8200 / 3 = 2733.33	2733 (remainder dropped by //)	Run on the source file
3	8000, 1000, 150, 9, 0	Error	ZeroDivisionError (no check for 0 persons)	Run on the source file
4	Text such as abc as rent	Error	ValueError (int() cannot convert text)	Run on the source file

Note: test case 1 is taken from the screenshot. Test cases 2 to 4 were executed on the same source file, Rent_calculator.py, to check how the program behaves on other inputs.

7. Conclusion

The Rent Calculator is a working Python program that takes the monthly costs of a shared room or flat and tells each person how much to pay. It gives the correct result for the sample values shown in the output screenshot and removes the need to calculate the share by hand.

What I learned

- Reading user input with input() and converting it with int().
- Using variables and arithmetic operators, including floor division (//).
- Writing a program in a clear order of input, processing and output.
- Testing a program with different inputs and noticing its limits.

Limitations and future improvements

- Add a check so that the number of persons cannot be zero and the inputs must be numbers (for example with try/except).
- Accept decimal values and show the exact share instead of dropping the remainder.
- Allow different shares for different persons, and show a breakdown of rent, food and electricity.
- Add a simple graphical interface and save the monthly records to a file.

8. GitHub Repository

Repository link: [Paste your GitHub repository link here, e.g. https://github.com/username/rent-calculator]

Source file: Rent_calculator.py. The comment at the top of the source file credits @codingwithsagar.